

Course Syllabus

Syllabus for CS 2505: Intro Computer Organization

Fall 2023

Instructors

- David McPherson
 - 1220A Torgersen Hall
 - dmcphers@vt.edu (<mailto:dmcphers@vt.edu>)
 - 231-3153
 - Office Hours:
Monday, Wednesday, and Friday, 9:30-11:30
- Siwei Cao
 - 2220B Torgersen Hall
 - csiwei@vt.edu (<mailto:csiwei@vt.edu>)
 - 231-5605
 - Office Hours: TR 2:00-3:30pm

Class Room Locations

CRN	Time	Room	Instructor
83324	TR 8:00-9:15am	NCB 160	McPherson
83325	TR 12:30-1:45pm	WHIT 300	Cao
83323	TR 2:00-3:15pm	TORG 2150	McPherson

Teaching Assistants

GTAs

Edmund Amanor	Samane Bayatkandi	Shruti Dongare	Ali Haisam Muhammad Rafid	Noah Provenzano
elamanor88	samanebk24	dshruti20	haisamrafid	noahpro

UTAs

Jordan Atanassov	Alexander Brown	Raunak Chitre	Sean Henderson	Deepika Reddy	Labiba Sajjad	Brinda Srivatsan	Uday Talasila	Tuan Minh Vuong Tran	Roman Zrajevsky
jordanatanassov	alexbrown26	raunakc24	hendersonsm25	deepikar	slabiba22	brindasrivatsan	udayt	minhvt22	romanz

TA Office Hours: Held in Torg 1230



[Link to Calendar](#) 

(<https://outlook.office365.com/owa/calendar/03826c4876184f848cf67d333deae4d3@vt.edu/c60ff04d44504ed58926995c69aad4a3904412667965349152/calendar.htm>)

Communication

To contact your instructors, please use email, **not** the Canvas message tool. We will use Piazza for our class forum. Please do **not** post source code, as this could be an honor violation. General questions about the class that only the class instructor can answer, please send those as emails. For examples, particular questions about the grading of your assignment.

Course Description

An introduction to high-level computer architecture and systems concepts, system software and programming in C.

Covers information representation, basic computer organization, fundamentals of the x86 architecture, the relationship between the C programming language and x86 assembly language, the use of debugging and other system tools, and the development of small- to medium-sized software solutions in C.

The course will focus on the x86-64 architecture.

Course Objectives

Having successfully completed this course, the student will be able to:

- understand the representation of information at the hardware level,
- understand the different layers of abstraction in a computing system, i.e., logic design, computer architecture, machine language, assembly language, high level language,
- write simple to moderately complex programs in C,
- read and understand simple programs in x86 assembly language.

- be able to apply common UNIX tools, such as `gdb` and `objdump`, to analyze, debug and correct C programs,
- be able to operate efficiently in a UNIX command-line environment.

Prerequisites/Corequisite

- A grade of C or better in CS 2114 (prerequisite)
- Math 2534, may be taken at the same time (corequisite)

Texts and Resources

Required Text

Computer Systems: a Programmer's Perspective, 3rd Edition

Randal E Bryant & David R O'Hallaron

Addison Wesley ©2016

ISBN 978-0-13-409266-9

[Text book website](https://csapp.cs.cmu.edu/)  (https://csapp.cs.cmu.edu/)

[Publisher's Website](https://www.pearson.com/en-us/subject-catalog/p/computer-systems-a-programmer-s-perspective/P200000003479/9780134154305)  (https://www.pearson.com/en-us/subject-catalog/p/computer-systems-a-programmer-s-perspective/P200000003479/9780134154305)

Recommended

A Practical Guide to Linux Commands, Editors, and Shell Programming, 2nd, 3rd or 4th Ed. Mark G Sobell

Prentice Hall ©2017

ISBN 978-0134774602 (4th Ed.)

[available via the Safari Database in the VT Library](https://learning.oreilly.com/library/view/a-practical-guide/9780134774626/)  (https://learning.oreilly.com/library/view/a-practical-guide/9780134774626/)

C Primer Plus, 6th Edition

Stephen Prata

Developer's Library ©2014

978-0-321-92842-9

[available via the VT Library](https://learning.oreilly.com/library/view/c-primer-plus/9780133432398/)  (https://learning.oreilly.com/library/view/c-primer-plus/9780133432398/)

–or–

Programming in C, 4th Edition

Stephen G Kochan

Developer's Library ©2014

978-0-321-77641-9

[Available via the VT Library](https://learning.oreilly.com/library/view/programming-in-c/9780132781206/)  (https://learning.oreilly.com/library/view/programming-in-c/9780132781206/)

Grading Scale

We will follow the standard 10 point scale. In particular that means that the following final average will guarantee the corresponding letter grade.

Average	Grade
93	A
90	A-
87	B+
83	B
80	B-
77	C+
73	C
70	C-
67	D+
63	D
60	D-
Below 60	F

Binary Bomb Requirement

Note: There is a requirement to pass at a minimum 4 phases of the binary bomb assignment. Failure to pass 4 phases will result in a 0 for the Binary Bomb Assignment.

Grade Distribution

We are planning on using this distribution for our assignments

Category	Weight
Homework	20%
Projects	40%
Tests (2)	20%
Final Exam	20%

Confirmation of assignment submission is the responsibility of the student. Please check that your assignments are submitted correctly.

Attendance

You are expected to be an active participant in class discussion and other learning opportunities. To do this, you must be prepared, so be sure to complete all reading and other assignments according to schedule. The class activities have been carefully designed to help you achieve the learning outcomes for the course. Missing class or not actively participating will negatively affect your ability to learn the content.

Late work

Late work is assessed a 10% per day penalty with a maximum deduction of 50%. Generally, there will be a 5 day maximum allowance to turn in work for credit; meaning work turned in more than 5 days late will not be graded. This will be enforced this semester.

Extension for Work

If you need an extension for an authorized excuse, please ask.

Simply having a lot of work to do, does not constitute a reason for an extension.

Being sick or some other sort of issue generally speaking does constitute a reason for an extension. You **must** provide some sort of documentation. If you are physically not well or are having a mental health issue, please get help and get a note.

Be aware that using an extension may negatively impact your ability to stay on track, so please only use them when absolutely needed.

Software

In this class, we are going to be using a container to access our compiler and related tools. To ease this connection we suggest using [VS Code](https://code.visualstudio.com/)  (<https://code.visualstudio.com/>). You can do this on the commandline if you wish. If you are on a Mac with the newer Apple Silicon chips, e.g. M1, M2, at some point later in the term you will need to connect to rlogin to be able to do the course work. More about that is coming. Please ask if you have any questions.

[Setting up your Development Environment \(https://canvas.vt.edu/courses/196087/pages/setting-up-your-development-environment\)](https://canvas.vt.edu/courses/196087/pages/setting-up-your-development-environment)

There are some pages on the CS Wiki about rlogin:

- [HowTo Page \(https://wiki.cs.vt.edu/wiki/HowTo\)](https://wiki.cs.vt.edu/wiki/HowTo)
- [HowTo rlogin \(https://wiki.cs.vt.edu/wiki/Howto::Access_rlogin_service\)](https://wiki.cs.vt.edu/wiki/Howto::Access_rlogin_service)
- [HowTo VSCode -> rlogin \(https://wiki.cs.vt.edu/wiki/Howto::VScode\)](https://wiki.cs.vt.edu/wiki/Howto::VScode)
- [HowTo Contact CS Techstaff \(https://wiki.cs.vt.edu/wiki/Contact_Techstaff\)](https://wiki.cs.vt.edu/wiki/Contact_Techstaff)

[Topics \(https://canvas.vt.edu/courses/196087/pages/topics\)](https://canvas.vt.edu/courses/196087/pages/topics)

Week	Date	Topic
1	Aug 26	• Course Policies and Overview • VSCode and rlogin • Linux: Basic Commands and Basic Scripting
2	Sep 2	• Navigating a Linux environment • Continued Basic Scripting • Beginning C (variables, data types, arrays, etc.)
3	Sept 9	• Compiling and Running C code (gcc)

Week	Date	Topic
		• Functions • Basic Output
4	Sept 16	• C Identifier Properties • Program Organization and Separate Compilation • Structs • Numeric Bases
5	Sept 23	• Debugging in C (gdb) • Dynamic Arrays • Intro to Pointers
6	Sept 30	• Strings • String I/O Parsing • Pointers and Arrays (valgrind)
	Oct 3	Drop Deadline
7	Oct 7	• Advanced Pointer Topics • Test 1
	Oct 11	Fall Break
8	Oct 14	• Bash Shell Scripting • Data Representation
9	Oct 21	• Bitwise Operations • Intro to Computer Architecture (von Neumann)
10	Oct 28	• Intro to x86-64 Assembly • Assembly Control Structures
11	Nov 4	• Functions and the Stack in Assembly
12	Nov 11	• Linking • More Assembly Examples
14	Nov 18	• Test 2 • Debugging Assembly
	Nov 20	Last Day to Reschedule Final Exam
	Nov 23	Thanksgiving Break
15	Dec 2	• Review
16	Dec 11	• Open Q + A
	Dec 13	Final Exam 4:25 pm-6:25 pm (Friday -- location TBD)

Final Exam

The Final Exam is a common-time exam. That means that all students taking CS2505 this semester will be taking the exam at the same time.

The exam will be held on **Friday, December 13 from 4:25 pm - 6:25 pm**.

Permission from the Dean's Office is necessary to take the final exam at a different time. If you have multiple exams at the same time or more than 2 exams within 24 hours, you must attempt to reschedule the **non-common-time** exams first.

If you have SSD accommodations for extended time/reduced distraction environment, an alternate time and place will be provided toward the end of the semester. If you have these accommodations, you **do not** need additional paperwork to take the exam at an alternate time.

The Virginia Tech Honor Code

The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states:

"As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do."

Students enrolled in this course are responsible for abiding by the [Honor Code \(https://www.honorsystem.vt.edu/\)](https://www.honorsystem.vt.edu/). A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Students are strongly encouraged to consult their faculty members regarding the use of any outside materials as the misuse of these sources may constitute a violation of the Honor Code. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code.

All assignments submitted shall be considered "graded work" and all aspects of your coursework are covered by the Honor Code. All projects and homework assignments are to be completed individually in this course unless otherwise specified. All written work must be written without help from other sources or people, except for the course instructor, the course TAs, and Student Success Center tutors. It is a violation of the Honor Code in this course to receive help from any other source, including online tutoring sites (including but not limited to Chegg, CourseHero, or GroupMe), or generative AI tools (including but not limited to ChatGPT, GitHub Copilot, and Microsoft Copilot).

The Academic Integrity expectations for Hokies are the same in an online class as they are in an in-person class. Hokies are expected to meet the academic integrity standards at Virginia Tech at all times.

Commission of any of the following acts shall constitute academic misconduct. This listing is not, however, exclusive of other acts that may reasonably be said to constitute academic misconduct. Clarification is provided for each definition with some examples of prohibited behaviors in the [Undergraduate Honor Code Manual \(https://www.honorsystem.vt.edu/\)](https://www.honorsystem.vt.edu/):

- **Cheating:** Cheating includes the intentional use of unauthorized materials, information, notes, study aids or other devices or materials in any academic exercise, or attempts thereof.
- **Plagiarism:** Plagiarism includes the copying of the language, structure, programming, computer code, ideas, and/or thoughts of another and passing off the same as one's own original work, or attempts thereof.
- **Falsification:** Falsification includes the statement of any untruth, either verbally or in writing, with respect to any element of one's academic work, or attempts thereof.
- **Fabrication:** Fabrication includes making up data and results, and recording or reporting them, or submitting fabricated documents, or attempts thereof.
- **Multiple Submission:** Multiple submission involves the submission for credit – without authorization from the instructor receiving the work – of substantial portions of any work (including oral reports) previously submitted for credit at any academic institution or attempts thereof.
- **Complicity:** Complicity includes intentionally helping another to engage in an act of academic misconduct, or attempts thereof.
- **Violation of University, College, Departmental, Program, Course, or Faculty Rules:** The violation of any University, College, Departmental, Program, Course, or Faculty Rules relating to academic matters that may lead to an unfair academic advantage by the student violating the rule(s).

Note that all electronic work submitted for this course is archived and subjected to automatic plagiarism detection and cheating analysis.

If you have questions or are unclear about what constitutes academic misconduct on an assignment, please speak with your instructor. We take the Honor Code seriously in this course. The normal sanction we will recommend for a violation of the Honor Code is an **F* sanction** as your final course grade. The F represents failure in the course. The identifies "*" a student who has failed to uphold the values of academic integrity at Virginia Tech. A student who receives a sanction of F* as their final course grade shall have it documented on their transcript with the notation "FAILURE DUE TO ACADEMIC HONOR CODE VIOLATION." You would be required to complete an education program administered by the Honor System in order to have the "*" and notation "FAILURE DUE TO ACADEMIC HONOR CODE VIOLATION" removed from your transcript. The "F" however would be permanently on your transcript.

Academic Accommodations

Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, [ssd@vt.edu \(mailto:ssd@vt.edu\)](mailto:ssd@vt.edu), or visit [https://ssd.vt.edu \(https://ssd.vt.edu\)](https://ssd.vt.edu)). If you have an SSD accommodation letter, please meet with me privately during office hours as early in the semester as possible to deliver your letter and discuss your accommodations. You must give me reasonable notice to implement your accommodations, which is generally 5 business days and 10 business days for final exams.

If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course.

Student Well Being

Supporting the mental health and well-being of students in my class is of high priority to me and Virginia Tech. If you are feeling overwhelmed academically, having trouble functioning, or are worried about a friend, please reach out to any of the following offices:

Cook Counseling

- Schedule an appointment and/or 24/7 crisis support: 540-231-6557

- <https://ucc.vt.edu> (<https://ucc.vt.edu>) for more information

Dean of Students Office

- General advice: 540 231-3787
- After-hours crisis: 540-231-6411
- <https://dos.vt.edu> (<https://dos.vt.edu>) for more information

Hokie Wellness

- <https://hokiewellness.vt.edu> (<https://hokiewellness.vt.edu>) for more information about health and wellness workshops and consultations
- [Virginia Tech Recovery Community](https://hokiewellness.vt.edu/students/recovery.html) (<https://hokiewellness.vt.edu/students/recovery.html>)

Services for Students with Disabilities (SSD)

- Accommodations and other disability-related supports: 540-231-3788
- <https://ssd.vt.edu> (<https://ssd.vt.edu>) for more information

See a full listing of campus resources on <https://well-being.vt.edu> (<https://well-being.vt.edu>).

Please also feel free to speak with me. I will make an effort to work with you; I care about your well-being and success.

Course Summary:

Date	Details	Due
Wed Sep 4, 2024	 Prior Knowledge Survey (https://canvas.vt.edu/courses/196087/assignments/2198015)	due by 11:59pm
Fri Sep 6, 2024	 Homework 1 (https://canvas.vt.edu/courses/196087/assignments/2198011)	due by 11:59pm
Sat Sep 7, 2024	 Week 2 Survey (https://canvas.vt.edu/courses/196087/assignments/2206186)	due by 11:59pm
Thu Sep 19, 2024	 Homework 2 (https://canvas.vt.edu/courses/196087/assignments/2198012)	due by 11:59pm
Fri Sep 27, 2024	 Project 1 (https://canvas.vt.edu/courses/196087/assignments/2214434)	due by 11:59pm
Fri Oct 4, 2024	 Homework 3 (https://canvas.vt.edu/courses/196087/assignments/2198014)	due by 11:59pm
Sat Oct 19, 2024	 Week 8 Survey (https://canvas.vt.edu/courses/196087/assignments/2206198)	due by 11:59pm
Mon Oct 21, 2024	 Project 2 (https://canvas.vt.edu/courses/196087/assignments/2214435)	due by 11:59pm
Mon Nov 4, 2024	 Homework 4 (https://canvas.vt.edu/courses/196087/assignments/2234377)	due by 11:59pm
Fri Nov 8, 2024	 Project 3: Datalab (https://canvas.vt.edu/courses/196087/assignments/2214436)	due by 11:59pm
Thu Nov 14, 2024	 Homework 5: Assembly Part 1 (https://canvas.vt.edu/courses/196087/assignments/2214423)	due by 11:59pm
	 Homework 5: Assembly Part 2 (https://canvas.vt.edu/courses/196087/assignments/2214424)	due by 11:59pm
	 Homework 5: Assembly Part 3 (https://canvas.vt.edu/courses/196087/assignments/2214425)	due by 11:59pm
	 Homework 5: Assembly Part 4 (https://canvas.vt.edu/courses/196087/assignments/2214426)	due by 11:59pm

Date	Details	Due
	 Homework 5: Total (https://canvas.vt.edu/courses/196087/assignments/2214427)	due by 11:59pm
Tue Nov 19, 2024	 Test 2 (https://canvas.vt.edu/courses/196087/assignments/2214440)	due by 11:59pm
Thu Nov 21, 2024	 Homework 6 (https://canvas.vt.edu/courses/196087/assignments/2214428)	due by 11:59pm
Fri Nov 29, 2024	 VT SPOT (https://canvas.vt.edu/calendar?event_id=1686183&include_contexts=course_196087)	12am
	 Homework 7: Part 1 (https://canvas.vt.edu/courses/196087/assignments/2214429)	due by 11:59pm
	 Homework 7: Part 2 (https://canvas.vt.edu/courses/196087/assignments/2214430)	due by 11:59pm
Wed Dec 4, 2024	 Homework 7: Part 3 (https://canvas.vt.edu/courses/196087/assignments/2214431)	due by 11:59pm
	 Homework 7: Part 4 (https://canvas.vt.edu/courses/196087/assignments/2214432)	due by 11:59pm
	 Homework 7: Total (https://canvas.vt.edu/courses/196087/assignments/2214433)	due by 11:59pm
Sun Dec 8, 2024	 Project 4 (https://canvas.vt.edu/courses/196087/assignments/2214437)	due by 11:59pm
Wed Dec 11, 2024	 Project 5: Bomblab (https://canvas.vt.edu/courses/196087/assignments/2214438)	due by 11:59pm
	 Week 15 Survey (https://canvas.vt.edu/courses/196087/assignments/2206203)	due by 11:59pm
Fri Dec 13, 2024	 Final Exam (https://canvas.vt.edu/courses/196087/assignments/2214422)	due by 6:25pm
	 Test 1 (https://canvas.vt.edu/courses/196087/assignments/2214439)	